

Geotechnical — pile, anchor & foundation capacity

Standards-referenced calculators for the seabed side of an offshore mooring or foundation problem — the engines that size and check the anchors the mooring-resilience atlases place. Pile axial capacity, drag- and suction-anchor holding, mudmat bearing and sliding, plus liquefaction triggering and scour screening. Each function is a dataclass-returning engine with input validation and its governing standard stamped on the result.

 α / β

clay / sand skin friction

3

seabed soil classes, live

4

engine families

WHAT YOU GET

- Driven-pile axial capacity per API RP 2GEO: α method in clay, β method in sand, end bearing, summed layer-by-layer
- Drag-anchor holding from empirical efficiency factors (DNV-RP-E302, API RP 2SK) live across soft clay / stiff clay / sand; suction-caisson capacity (DNV-RP-E303) in the same engine
- Mudmat bearing capacity (DNV-RP-C212 / Brinch Hansen) with shape, depth and effective-area corrections, plus base sliding
- Seed-Idriss / NCEER per-layer liquefaction factor of safety with magnitude scaling
- DNV-RP-F107 equilibrium scour for pipelines and monopiles and rock-armour thickness